

Decimal Operations

Every graded set, every week — 5 weeks, 25 two-page sheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

WEEK 5

5 sets

HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

Adding Decimals

Line up the points, fill with zeros, then add as normal.

Warm-Up 12 ITEMS • RECALL

1 $0.3 + 0.4$

2 $1.2 + 2.5$

3 $0.25 + 0.25$

4 $3.5 + 1.5$

5 $0.6 + 0.9$

6 $2.75 + 1.25$

7 $0.5 + 0.2$

8 $1.4 + 2.3$

9 $0.35 + 0.15$

10 $3.5 + 2.5$

11 $0.8 + 0.7$

12 $4.2 + 3.1$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $3.5 + 0.47$

14. $12.6 + 4.85$

15. $0.125 + 0.875$

16. $9.4 + 0.68$

17. $\$14.75 + \8.60 — type the number

18. $4.6 + 0.38$

19. $15.4 + 3.75$

20. $8.3 + 0.79$

21. $0.625 + 0.375$

22. $6.04 + 3.8$



Working space



Error journal

Subtracting Decimals

Same alignment. Fill the gaps with zeros before you borrow.

Warm-Up 12 ITEMS • RECALL

1 $0.9 - 0.4$

2 $3.7 - 1.2$

3 $1 - 0.5$

4 $2.5 - 0.5$

5 $0.75 - 0.25$

6 $5.5 - 2.5$

7 $0.8 - 0.5$

8 $4.7 - 2.2$

9 $1 - 0.4$

10 $3.5 - 1.5$

11 $0.85 - 0.35$

12 $6.9 - 3.4$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $4 - 1.35$

14. $10 - 0.07$

15. $8.2 - 3.45$

16. $\$20 - \13.68 — type the number

17. $6.05 - 2.5$

18. $5 - 2.45$

19. $10 - 0.09$

20. $9.2 - 4.55$

21. $7.05 - 3.5$

22. $13.4 - 5.85$

Multiplying Decimals

Multiply the digits, then count the decimal places.

Warm-Up 12 ITEMS • RECALL

1 0.5×4

2 0.2×6

3 1.5×2

4 0.25×4

5 3×0.1

6 0.5×10

7 0.5×6

8 0.2×8

9 1.5×4

10 0.25×8

11 2.5×2

12 0.1×9

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. 0.4×0.2 (Two decimal digits in, two out)

14. 1.2×0.5

15. 2.5×1.4

16. 0.06×40

17. 3.2×2.5

18. 0.3×0.4

19. 1.5×0.6

20. 3.5×1.2

21. 0.08×50

22. 2.4×3.5

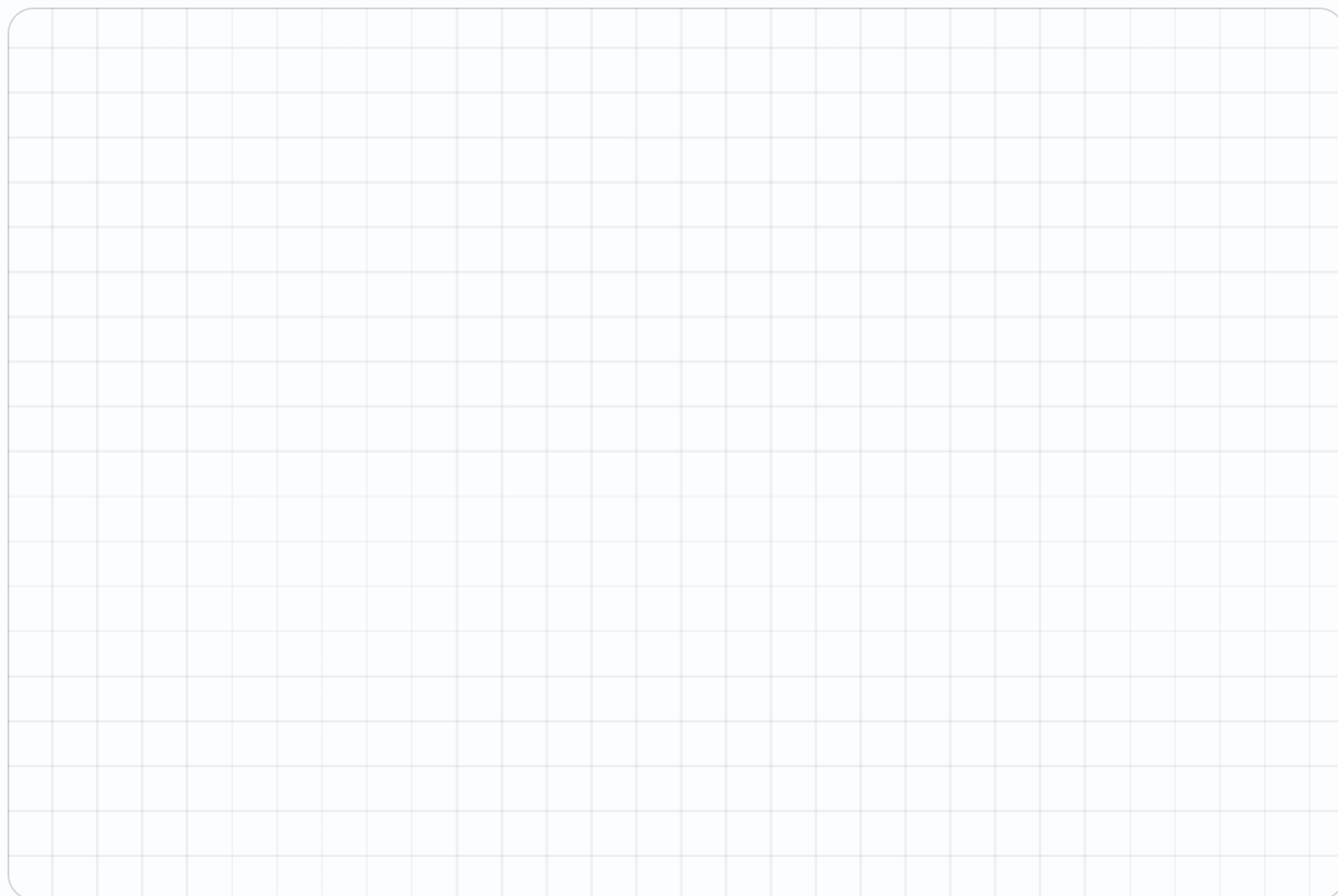
★ Challenge NOT GRADED • REASON IT OUT

23. 0.125×8
.....

24. 1.25×0.8
.....

25. 0.375×8
.....

26. 1.25×2.4
.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Dividing Decimals

Shift both numbers the same number of places, then divide as usual.

Warm-Up 12 ITEMS • RECALL

1 $4.8 \div 2$

2 $0.9 \div 3$

3 $6.4 \div 4$

4 $2.5 \div 5$

5 $0.36 \div 6$

6 $7.2 \div 8$

7 $6.4 \div 2$

8 $0.8 \div 4$

9 $4.5 \div 5$

10 $8 \div 0.5$

11 $1.2 \div 3$

12 $6 \div 0.25$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $6 \div 0.5$ (How many halves in six)

14. $4 \div 0.25$

15. $1.5 \div 0.5$

16. $9.6 \div 1.2$

17. \$7.50 shared by 3 — type the number

18. $9.6 \div 8$

19. $1.28 \div 4$

20. $12 \div 0.4$

21. $7.2 \div 0.9$

22. $0.96 \div 1.2$

Unit Price Investigation

Enrichment. Price per unit is the only fair comparison.

✦ Warm-Up 4 ITEMS • RECALL

1 \$6 for 2 kg — price per kg

2 \$10 for 5 L — price per litre

3 \$9 for 3 — each

4 \$14 for 7 — each

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. \$4.50 for 3 kg — price per kg

.....

6. \$7.20 for 0.8 kg — price per kg

.....

7. \$5.40 for 4 — each

.....

8. \$8.70 for 6 — each

.....

Mark the reasoning, not the handwriting.

Week 1 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

1.1 • Adding Decimals

OUT OF 22

Warm-Up • 1-12 • 0.7 | 3.7 | 0.5 | 5 | 1.5 | 4 | 0.7 | 3.7 | 0.5 | 6 | 1.5 | 7.3

Core • 13-22 • 3.97 | 17.45 | 1 | 10.08 | 23.35 | 4.98 | 19.15 | 9.09 | 1 | 9.84

Challenge • 23-26 • 6.375 | 5.4 | 7.875 | 6.3

1.2 • Subtracting Decimals

OUT OF 22

Warm-Up • 1-12 • 0.5 | 2.5 | 0.5 | 2 | 0.5 | 3 | 0.3 | 2.5 | 0.6 | 2 | 0.5 | 3.5

Core • 13-22 • 2.65 | 9.93 | 4.75 | 6.32 | 3.55 | 2.55 | 9.91 | 4.65 | 3.55 | 7.55

Challenge • 23-26 • 4.544 | 1.65 | 5.554 | 2.35

1.3 • Multiplying Decimals

OUT OF 22

Warm-Up • 1-12 • 2 | 1.2 | 3 | 1 | 0.3 | 5 | 3 | 1.6 | 6 | 2 | 5 | 0.9

Core • 13-22 • 0.08 | 0.6 | 3.5 | 2.4 | 8 | 0.12 | 0.9 | 4.2 | 4 | 8.4

Challenge • 23-26 • 1 | 1 | 3 | 3

1.4 • Dividing Decimals

OUT OF 22

Warm-Up • 1-12 • 2.4 | 0.3 | 1.6 | 0.5 | 0.06 | 0.9 | 3.2 | 0.2 | 0.9 | 16 | 0.4 | 24

Core • 13-22 • 12 | 16 | 3 | 8 | 2.5 | 1.2 | 0.32 | 30 | 8 | 0.8

Challenge • 23-26 • 1.2 | 6 | 3.5 | 125

Fri • Unit Price Investigation

OUT OF 8

Warm-Up • 1-4 • 3 | 2 | 3 | 2

Core • 5-8 • 1.5 | 9 | 1.35 | 1.45

Challenge • 9-14 • 1.125 | 1.25 | B | 1.2 | 1.1 | 1kg

Ignore the Point

Multiply the digits first, then count the places.

✦ Warm-Up 12 ITEMS • RECALL

1 5×4

2 0.5×4

3 12×5

4 1.2×5

5 25×4

6 0.25×4

7 6×3

8 0.6×3

9 14×5

10 1.4×5

11 35×4

12 0.35×4

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. 0.4×0.2

14. 1.2×0.5

15. 2.5×1.4

16. 0.06×40

17. 3.2×2.5

18. 0.6×0.3

19. 1.4×0.5

20. 3.5×1.6

21. 0.07×30

22. 4.2×2.5

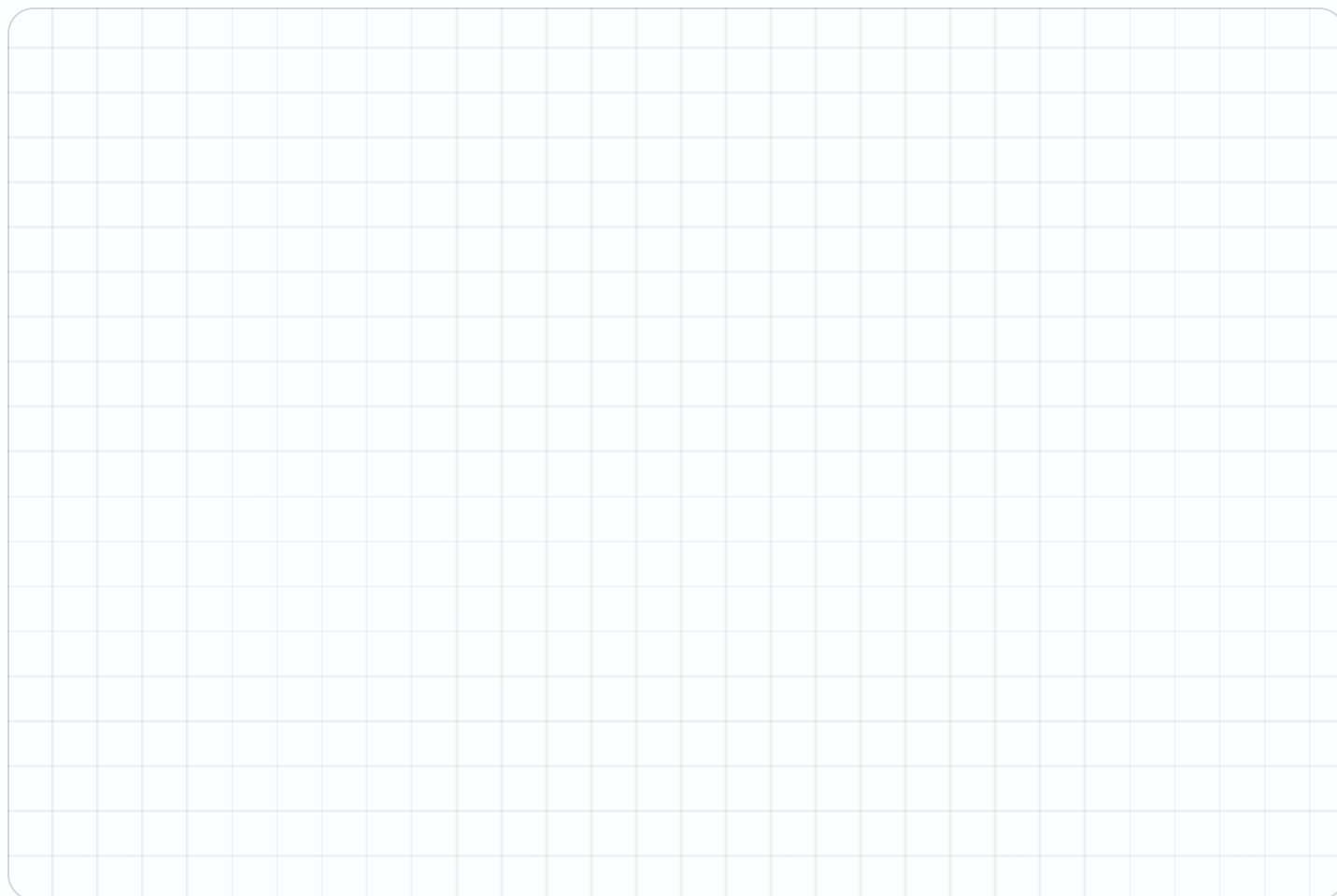
★ Challenge NOT GRADED • REASON IT OUT

23. 0.125×8
.....

24. 1.25×0.8
.....

25. 0.375×16
.....

26. 2.25×0.8
.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Count the Places

Two decimal digits in, two decimal digits out.

Warm-Up 12 ITEMS • RECALL

1 Decimal digits in 0.4

2 In 0.25

3 In 1.234

4 0.2×0.3

5 0.5×0.2

6 0.1×0.1

7 Decimal digits in 0.06

8 In 3.5

9 In 0.4567

10 0.3×0.3

11 0.4×0.5

12 0.2×0.2

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. 0.25×0.4

.....

14. 1.5×0.02

.....

15. 0.12×0.5

.....

16. 2.4×0.25

.....

17. 0.75×0.8

.....

18. 0.35×0.2

.....

19. 2.5×0.04

.....

20. 0.14×0.5

.....

21. 3.6×0.25

.....

22. 0.85×0.4

.....

Place It by Estimating

0.4×60 is about 24, so the point can only go one place.

Warm-Up 12 ITEMS • RECALL

1 Estimate 0.5×40

2 0.5×40

3 Estimate 2×30

4 1.9×30

5 Estimate 0.25×80

6 0.25×80

7 Estimate 0.5×60

8 0.5×60

9 Estimate 3×20

10 2.9×20

11 Estimate 0.25×40

12 0.25×40

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Estimate 0.4×60

.....

14. 0.4×62

.....

15. Estimate 4.9×21

.....

16. 4.9×20

.....

17. Estimate 0.9×150

.....

18. Estimate 0.6×80

.....

19. 0.6×82

.....

20. Estimate 3.9×31

.....

21. 3.9×30

.....

22. Estimate 0.8×250

.....

Smaller Than You Started

Why $\times 0.5$ halves instead of doubling.

Warm-Up 12 ITEMS • RECALL

1 10×0.5

2 10×2

3 10×1

4 8×0.5

5 20×0.25

6 6×0.5

7 20×0.5

8 20×2

9 12×0.5

10 40×0.25

11 20×1

12 16×0.5

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 40×0.75 — bigger or smaller than 40

14. 40×0.75

15. 40×1.25 — bigger or smaller

16. 40×1.25

17. 100×0.1

18. 60×0.75 — bigger or smaller than 60

19. 60×0.75

20. 60×1.5 — bigger or smaller

21. 60×1.5

22. 200×0.1



Point Placement

Given the digits, race to place the decimal point.

✦ Warm-Up 4 ITEMS • RECALL

1 Digits 24, one decimal place — the number

2 Digits 24, two places

3 Digits 36, one decimal place — the number

4 Digits 36, two places

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. 0.6×0.4 — digits 24, so the answer

.....

6. 6×0.4 — the answer

.....

7. 0.9×0.4 — digits 36, so the answer

.....

8. 9×0.4 — the answer

.....

Challenge NOT GRADED • REASON IT OUT

9. 1.5×0.16 — digits 240, so the answer

10. 0.15×1.6

11. 15×0.016

12. 1.2×0.3 — the answer

13. 0.12×3

14. 12×0.03

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Week 2 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

2.1 • Ignore the Point

OUT OF 22

Warm-Up • 1-12 • 20 | 2 | 60 | 6 | 100 | 1 | 18 | 1.8 | 70 | 7 | 140 | 1.4

Core • 13-22 • 0.08 | 0.6 | 3.5 | 2.4 | 8 | 0.18 | 0.7 | 5.6 | 2.1 | 10.5

Challenge • 23-26 • 1 | 1 | 6 | 1.8

2.2 • Count the Places

OUT OF 22

Warm-Up • 1-12 • 1 | 2 | 3 | 0.06 | 0.1 | 0.01 | 2 | 1 | 4 | 0.09 | 0.2 | 0.04

Core • 13-22 • 0.1 | 0.03 | 0.06 | 0.6 | 0.6 | 0.07 | 0.1 | 0.07 | 0.9 | 0.34

Challenge • 23-26 • 0.05 | 2 | 0.3 | 2.8

2.3 • Place It by Estimating

OUT OF 22

Warm-Up • 1-12 • 20 | 20 | 60 | 57 | 20 | 20 | 30 | 30 | 60 | 58 | 10 | 10

Core • 13-22 • 24 | 24.8 | 100 | 98 | 135 | 48 | 49.2 | 120 | 117 | 200

Challenge • 23-26 • 180 | 500 | 240 | 900

2.4 • Smaller Than You Started

OUT OF 22

Warm-Up • 1-12 • 5 | 20 | 10 | 4 | 5 | 3 | 10 | 40 | 6 | 10 | 20 | 8

Core • 13-22 • smaller | 30 | bigger | 50 | 10 | smaller | 45 | bigger | 90 | 20

Challenge • 23-26 • smaller | 3 | same | 4

Fri • Point Placement

OUT OF 8

Warm-Up • 1-4 • 2.4 | 0.24 | 3.6 | 0.36

Core • 5-8 • 0.24 | 2.4 | 0.36 | 3.6

Challenge • 9-14 • 0.24 | 0.24 | 0.24 | 0.36 | 0.36 | 0.36

Decimal ÷ Whole

The point comes straight up. Nothing else moves.

✦ Warm-Up 12 ITEMS • RECALL

1 $4.8 \div 2$

2 $0.9 \div 3$

3 $6.4 \div 4$

4 $2.5 \div 5$

5 $0.36 \div 6$

6 $7.2 \div 8$

7 $6.9 \div 3$

8 $0.8 \div 2$

9 $8.4 \div 4$

10 $3.5 \div 7$

11 $0.48 \div 8$

12 $5.4 \div 9$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $9.6 \div 4$

14. $1.44 \div 12$

15. $5.25 \div 5$

16. $8.4 \div 7$

17. $0.72 \div 9$

18. $1.32 \div 11$

19. $6.35 \div 5$

20. $9.8 \div 7$

21. $0.63 \div 9$

Whole ÷ Decimal

Shift both the same number of places, then divide as usual.

Warm-Up 11 ITEMS • RECALL

1 $6 \div 0.5$

2 $4 \div 0.5$

3 $10 \div 0.5$

4 $4 \div 0.25$

5 $3 \div 0.5$

6 $2 \div 0.25$

7 $8 \div 0.5$

8 $20 \div 0.5$

9 $8 \div 0.25$

10 $5 \div 0.5$

11 $3 \div 0.25$

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. $9 \div 0.3$

13. $12 \div 0.4$

14. $5 \div 0.125$

15. $7 \div 0.7$

16. $20 \div 0.8$

17. $12 \div 0.3$

18. $18 \div 0.6$

19. $6 \div 0.125$

20. $9 \div 0.9$

21. $30 \div 0.8$

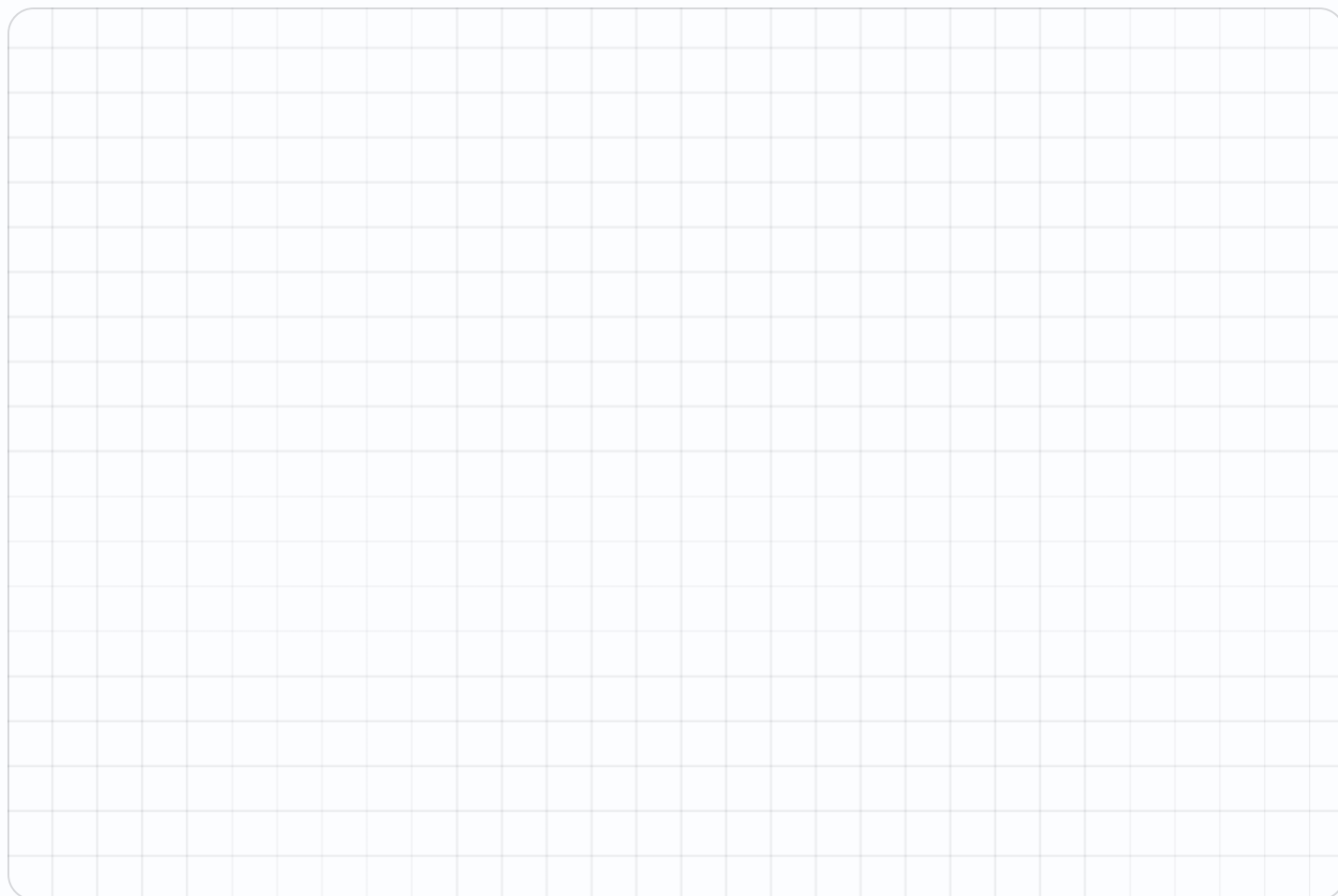
★ Challenge NOT GRADED • REASON IT OUT

22. $15 \div 0.06$
.....

23. $1 \div 0.004$
.....

24. $21 \div 0.07$
.....

25. $2 \div 0.004$
.....

✎ Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE**! Error journal** ONE SENTENCE PER MISTAKE

Decimal ÷ Decimal

Same shift, applied to both numbers.

✦ Warm-Up 11 ITEMS • RECALL

1 $0.6 \div 0.2$

2 $0.8 \div 0.4$

3 $1.5 \div 0.5$

4 $0.9 \div 0.3$

5 $2.4 \div 0.6$

6 $1.2 \div 0.4$

7 $1.2 \div 0.6$

8 $2.5 \div 0.5$

9 $1.8 \div 0.3$

10 $3.6 \div 0.9$

11 $1.6 \div 0.4$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. $9.6 \div 1.2$

13. $0.144 \div 0.12$

14. $4.5 \div 1.5$

15. $6.25 \div 2.5$

16. $0.81 \div 0.09$

17. $8.4 \div 1.2$

18. $0.288 \div 0.12$

19. $6.3 \div 1.5$

20. $7.5 \div 2.5$

21. $0.64 \div 0.08$



Challenge

22. $7.5 \div 0.25$

23. $0.0144 \div 0.012$

24. $9.6 \div 0.32$

25. $0.0288 \div 0.012$



Working space



Error journal

Money Problems

Unit prices and change, to the cent.

Warm-Up 12 ITEMS • RECALL

1 \$6 for 2 — each

2 \$10 for 4 — each

3 \$9 for 3 — each

4 \$20 — \$12.50

5 \$5 ÷ 4

6 3 at \$2.50

7 \$8 for 2 — each

8 \$12 for 4 — each

9 \$15 for 3 — each

10 \$20 — \$14.50

11 \$6 ÷ 4

12 4 at \$2.25

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. \$7.50 shared by 3

.....

14. \$4.50 for 3 kg — per kg

.....

15. \$7.20 for 0.8 kg — per kg

.....

16. \$20 — 3 at \$4.99

.....

17. \$15 ÷ 12

.....

18. \$9.60 shared by 4

.....

19. \$6.75 for 3 kg — per kg

.....

20. \$8.40 for 0.6 kg — per kg

.....

21. \$50 — 4 at \$6.99

.....

22. \$18 ÷ 12

.....



Working space



Error journal

Unit Price Investigation Begins

Collect real prices and sizes. No conclusions yet.

✦ Warm-Up 4 ITEMS • RECALL

1 \$8 for 4 — each

2 \$12 for 6 — each

3 \$10 for 5 — each

4 \$18 for 6 — each

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Short Core today — do all of them, then turn the page over.

5. \$3.60 for 4 — each

.....

6. \$11.25 for 5 — each

.....

7. \$4.80 for 4 — each

.....

8. \$13.75 for 5 — each

.....

Challenge NOT GRADED • REASON IT OUT

9. 1 kg at \$8, or 750 g at \$5.40 — per kg for the small one

10. Which is better value — type 1kg or 750g

11. \$2.40 for 6 — each

12. 2 kg at \$15, or 1.5 kg at \$12 — per kg for the small one

13. Which is better value — type 2kg or 1.5kg

14. \$3.60 for 8 — each

 Working space SHOW THE ROOMS, THE GRID, THE NUMBER LINE

Error journal ONE SENTENCE PER MISTAKE

Week 3 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

3.1 • Decimal ÷ Whole

OUT OF 21

Warm-Up • 1-12 • 2.4 | 0.3 | 1.6 | 0.5 | 0.06 | 0.9 | 2.3 | 0.4 | 2.1 | 0.5 | 0.06
| 0.6

Core • 13-21 • 2.4 | 0.12 | 1.05 | 1.2 | 0.08 | 0.12 | 1.27 | 1.4 | 0.07

Challenge • 22-25 • 3.125 | 0.49 | 3.625 | 0.63

3.2 • Whole ÷ Decimal

OUT OF 21

Warm-Up • 1-11 • 12 | 8 | 20 | 16 | 6 | 8 | 16 | 40 | 32 | 10 | 12

Core • 12-21 • 30 | 30 | 40 | 10 | 25 | 40 | 30 | 48 | 10 | 37.5

Challenge • 22-25 • 250 | 250 | 300 | 500

3.3 • Decimal ÷ Decimal

OUT OF 21

Warm-Up • 1-11 • 3 | 2 | 3 | 3 | 4 | 3 | 2 | 5 | 6 | 4 | 4

Core • 12-21 • 8 | 1.2 | 3 | 2.5 | 9 | 7 | 2.4 | 4.2 | 3 | 8

Challenge • 22-25 • 30 | 1.2 | 30 | 2.4

3.4 • Money Problems

OUT OF 22

Warm-Up • 1-12 • 3 | 2.5 | 3 | 7.5 | 1.25 | 7.5 | 4 | 3 | 5 | 5.5 | 1.5 | 9

Core • 13-22 • 2.5 | 1.5 | 9 | 5.03 | 1.25 | 2.4 | 2.25 | 14 | 22.04 | 1.5

Challenge • 23-26 • 1.25 | 1.125 | 1.2 | 1.1

Fri • Unit Price Investigation Begins

OUT OF 8

Warm-Up • 1-4 • 2 | 2 | 2 | 3

Core • 5-8 • 0.9 | 2.25 | 1.2 | 2.75

Challenge • 9-14 • 7.2 | 750g | 0.4 | 8 | 2kg | 0.45

Choose the Operation

The words tell you. Read them twice before reaching for a method.

Warm-Up 12 ITEMS • RECALL

1 3 items at \$2.50 — total

2 \$10 shared by 4

3 \$5.50 + \$2.50

4 \$10 – \$3.25

5 4 at \$1.25

6 \$9 ÷ 3

7 4 items at \$3.50 — total

8 \$12 shared by 3

9 \$6.50 + \$3.50

10 \$20 – \$4.75

11 5 at \$1.40

12 \$18 ÷ 6

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 2.5 kg at \$4 per kg

.....

14. \$12 buys how many at \$2.40

.....

15. 1.5 m plus 0.85 m — metres

.....

16. 6 at \$3.99 — total

.....

17. \$50 – \$27.45

.....

18. 3.5 kg at \$6 per kg

.....

19. \$15 buys how many at \$2.50

.....

20. 2.4 m plus 1.35 m — metres

.....

21. 7 at \$2.99 — total

.....

22. \$80 – \$34.65

.....

Two-Step Problems

Multiply then subtract, in that order.

Warm-Up 12 ITEMS • RECALL

1 3 at \$2, change from \$10

2 4 at \$1.50, change from \$10

3 2 at \$3.25 — total

4 2 at \$3.25 — change from \$10

5 5 at \$2 — total

6 5 at \$2 — change from \$20

7 4 at \$2, change from \$10

8 3 at \$1.50, change from \$10

9 3 at \$2.25 — total

10 3 at \$2.25 — change from \$10

11 6 at \$2 — total

12 6 at \$2 — change from \$20

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. 3 at \$4.99, change from \$20

14. 6 at \$1.75 — total

15. 6 at \$1.75 — change from \$20

16. 2.5 kg at \$3.20, change from \$10

17. 4 at \$6.25 — total

18. 4 at \$3.99, change from \$20

19. 5 at \$1.85 — total

20. 5 at \$1.85 — change from \$20

21. 3.5 kg at \$2.40, change from \$10

22. 3 at \$8.25 — total

Measurement Contexts

Metres, litres and kilograms, all carrying decimals.

✦ Warm-Up 12 ITEMS • RECALL

1 $1.5 \text{ m} + 0.5 \text{ m}$

2 $2.5 \text{ L} - 1 \text{ L}$

3 $0.5 \text{ kg} \times 4$

4 $3 \text{ m} \div 2$

5 $1.2 \text{ m} + 0.8 \text{ m}$

6 $4.5 \text{ kg} - 1.5 \text{ kg}$

7 $2.5 \text{ m} + 1.5 \text{ m}$

8 $3.5 \text{ L} - 1.5 \text{ L}$

9 $0.25 \text{ kg} \times 8$

10 $5 \text{ m} \div 2$

11 $1.6 \text{ m} + 0.4 \text{ m}$

12 $6.5 \text{ kg} - 2.5 \text{ kg}$

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. $1.25 \text{ m} \times 4$

14. 7.5 L shared into 0.5 L bottles — how many

15. $2.4 \text{ kg} \div 3$

16. $0.75 \text{ m} \times 6$

17. $12.5 \text{ L} - 4.75 \text{ L}$

18. $1.75 \text{ m} \times 4$

19. 9 L shared into 0.75 L bottles — how many

20. $3.6 \text{ kg} \div 4$

21. $0.45 \text{ m} \times 8$

22. $20.5 \text{ L} - 8.75 \text{ L}$



Working space



Error journal

Estimate Everything First

Write the estimate before the working, every time.

Warm-Up 12 ITEMS • RECALL

1 Estimate $4.9 + 3.1$

2 Estimate $9.8 - 4.9$

3 Estimate 2.1×4

4 Estimate $11.9 \div 4$

5 Estimate $0.9 + 1.1$

6 Estimate 5.2×2

7 Estimate $5.9 + 4.1$

8 Estimate $10.2 - 5.1$

9 Estimate 3.1×3

10 Estimate $15.9 \div 4$

11 Estimate $1.9 + 2.1$

12 Estimate 6.1×2

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. Estimate $12.4 - 7.9$

.....

14. True value of $12.4 - 7.856$

.....

15. Estimate 0.49×100

.....

16. True value of 0.49×100

.....

17. Estimate $19.8 \div 5$

.....

18. Estimate $14.6 - 8.9$

.....

19. True value of $14.6 - 8.746$

.....

20. Estimate 0.51×200

.....

21. True value of 0.51×200

.....

22. Estimate $24.6 \div 5$

.....



Mid-Unit Quiz

Eight items across Weeks 1–4. 85% to keep flying.

✦ Warm-Up 4 ITEMS • RECALL

1 $0.3 + 0.4$

2 0.5×4

3 $0.5 + 0.2$

4 0.6×3

◆ Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. $3.5 + 0.47$

6. $8.2 - 3.45$

7. 0.4×0.2

8. $6 \div 0.5$

9. \$7.20 for 0.8 kg — per kg

10. $4.6 + 0.38$

11. $9.2 - 4.55$

12. 0.6×0.3

13. $8 \div 0.5$

14. \$8.40 for 0.6 kg — per kg

Week 4 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

4.1 • Choose the Operation

OUT OF 22

Warm-Up • 1-12 • 7.5 | 2.5 | 8 | 6.75 | 5 | 3 | 14 | 4 | 10 | 15.25 | 7 | 3

Core • 13-22 • 10 | 5 | 2.35 | 23.94 | 22.55 | 21 | 6 | 3.75 | 20.93 | 45.35

Challenge • 23-26 • 4.8 | 24 | 9 | 60

4.2 • Two-Step Problems

OUT OF 22

Warm-Up • 1-12 • 4 | 4 | 6.5 | 3.5 | 10 | 10 | 2 | 5.5 | 6.75 | 3.25 | 12 | 8

Core • 13-22 • 5.03 | 10.5 | 9.5 | 2 | 25 | 4.04 | 9.25 | 10.75 | 1.6 | 24.75

Challenge • 23-26 • 6.25 | 9.7 | 9 | 17.5

4.3 • Measurement Contexts

OUT OF 22

Warm-Up • 1-12 • 2 | 1.5 | 2 | 1.5 | 2 | 3 | 4 | 2 | 2 | 2.5 | 2 | 4

Core • 13-22 • 5 | 15 | 0.8 | 4.5 | 7.75 | 7 | 12 | 0.9 | 3.6 | 11.75

Challenge • 23-26 • 6 | 0.45 | 7 | 0.45

4.4 • Estimate Everything First

OUT OF 22

Warm-Up • 1-12 • 8 | 5 | 8 | 3 | 2 | 10 | 10 | 5 | 9 | 4 | 4 | 12

Core • 13-22 • 4 | 4.544 | 50 | 49 | 4 | 6 | 5.854 | 100 | 102 | 5

Challenge • 23-26 • 30 | 29.7 | 32 | 31.6

Fri • Mid-Unit Quiz

OUT OF 14

Warm-Up • 1-4 • 0.7 | 2 | 0.7 | 1.8

Core • 5-14 • 3.97 | 4.75 | 0.08 | 12 | 9 | 4.98 | 4.65 | 0.18 | 16 | 14

Challenge • 15-16 • 4.544 | 5.554

Compute Unit Prices

Divide price by size for every item on the list.

Warm-Up 12 ITEMS • RECALL

1 \$6 for 2 kg — per kg

2 \$10 for 5 L — per L

3 \$8 for 4 — each

4 \$12 for 3 — each

5 \$15 for 5 — each

6 \$9 for 3 — each

7 \$8 for 2 kg — per kg

8 \$12 for 6 L — per L

9 \$10 for 5 — each

10 \$16 for 4 — each

11 \$18 for 6 — each

12 \$21 for 7 — each

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. \$4.50 for 3 kg — per kg

.....

14. \$7.20 for 0.8 kg — per kg

.....

15. \$11.25 for 5 — each

.....

16. \$2.40 for 6 — each

.....

17. \$13.50 for 1.5 kg — per kg

.....

18. \$6.75 for 3 kg — per kg

.....

19. \$8.40 for 0.6 kg — per kg

.....

20. \$13.75 for 5 — each

.....

21. \$3.60 for 8 — each

.....

22. \$16.20 for 1.8 kg — per kg

.....



Challenge

NOT GRADED • REASON IT OUT

23. \$5 for 400 g — per 100 g

24. \$9 for 800 g — per 100 g

25. \$6 for 500 g — per 100 g

26. \$11 for 1 kg — per 100 g



Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE



Error journal

ONE SENTENCE PER MISTAKE

Find the Trap

Name one product where the bigger box costs more per unit.

Warm-Up 10 ITEMS • RECALL

1 \$4 for 2 — each

2 \$9 for 5 — each

3 Cheaper each — type 1.8 or 2

4 \$6 for 3 — each

5 \$10 for 4 — each

6 Cheaper each — type 2 or 2.5

7 \$10 for 6 — each, to the cent

8 Cheaper each — type 1.67 or 2

9 \$8 for 4 — each

10 \$15 for 6 — each

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. 400 g at \$5 — per 100 g

.....

12. 800 g at \$9 — per 100 g

.....

13. Better value — type 400 or 800

.....

14. 500 g at \$4 — per 100 g

.....

15. 1 kg at \$9 — per 100 g

.....

16. 500 g at \$6 — per 100 g

.....

17. 1 kg at \$11 — per 100 g

.....

18. Better value — type 500 or 1000

.....

19. 250 g at \$3 — per 100 g

.....

20. 750 g at \$8.25 — per 100 g

.....

Challenge

21. So which is better value — type 500g or 1kg

22. The saving per 100 g by picking it

23. So which is better value — type 250g or 750g



Working space



Error journal

Write the Recommendation

One paragraph a shopper could act on.

Warm-Up 12 ITEMS • RECALL

1 \$1.25 per 100 g — the cost of 400 g

2 \$0.90 per 100 g — the cost of 1 kg

3 \$2 each — the cost of 5

4 \$1.80 each — the cost of 5

5 The saving

6 \$3 per kg — the cost of 2 kg

7 \$1.20 per 100 g — the cost of 500 g

8 \$1.10 per 100 g — the cost of 1 kg

9 \$2 each — the cost of 6

10 \$1.67 each — the cost of 6, to the cent

11 \$3 per kg — the cost of 3 kg

12 \$4 each — the cost of 4

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Saving \$0.125 per 100 g on 800 g

.....

14. Buying 800 g weekly — the yearly saving over 52 weeks

.....

15. \$1.25 vs \$1.125 per 100 g — the percent saved, to the nearest whole

.....

16. \$9 vs \$10 — the percent saved

.....

17. Saving \$1 a week for a year

.....

18. Saving \$0.10 per 100 g on 1 kg

.....

19. Buying 1 kg weekly — the yearly saving over 52 weeks

.....

20. \$1.20 vs \$1.10 per 100 g — the percent saved, to the nearest whole

.....

21. \$11 vs \$12 — the percent saved, to the nearest whole

.....

22. Saving \$2 a week for a year

.....

Error Journal Sweep

Re-read every entry from the mission. Fix only what repeats.

Warm-Up 12 ITEMS • RECALL

1 $0.25 + 0.25$

2 $1 - 0.5$

3 0.5×4

4 $4.8 \div 2$

5 $0.6 + 0.9$

6 \$10 for 5 — each

7 $0.35 + 0.15$

8 $1 - 0.4$

9 0.2×8

10 $6.9 \div 3$

11 $0.8 + 0.7$

12 \$12 for 4 — each

Core GRADED • SHOW YOUR WORKING

STOP RULE Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13. $12.6 + 4.85$

14. $10 - 0.07$

15. 0.4×0.2

16. $9.6 \div 1.2$

17. 2.5×1.4

18. $15.4 + 3.75$

19. $10 - 0.09$

20. 0.6×0.3

21. $8.4 \div 1.2$

22. 3.5×1.6

Mission 04 Test

Twelve items plus the Big Question, answered out loud.

Warm-Up 4 ITEMS • RECALL

1 $1.2 + 2.5$

2 $0.9 - 0.4$

3 $1.4 + 2.3$

4 $0.8 - 0.5$

Core GRADED • SHOW YOUR WORKING

STOP RULE This one is a test. Do **every** problem — no stopping early.

5. $3.5 + 0.47$

6. $4 - 1.35$

7. 1.2×0.5

8. 3.2×2.5

9. $6 \div 0.5$

10. $9.6 \div 1.2$

11. \$4.50 for 3 kg — per kg

12. 3 at \$4.99 — change from \$20

13. $4.6 + 0.38$

14. $5 - 2.45$

15. 1.4×0.5

16. 4.2×2.5

17. $8 \div 0.5$

Week 5 Answers

Mark the reasoning, not the handwriting.

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

5.1 • Compute Unit Prices

OUT OF 22

Warm-Up • 1-12 • 3 | 2 | 2 | 4 | 3 | 3 | 4 | 2 | 2 | 4 | 3 | 3

Core • 13-22 • 1.5 | 9 | 2.25 | 0.4 | 9 | 2.25 | 14 | 2.75 | 0.45 | 9

Challenge • 23-26 • 1.25 | 1.125 | 1.2 | 1.1

5.2 • Find the Trap

OUT OF 20

Warm-Up • 1-10 • 2 | 1.8 | 1.8 | 2 | 2.5 | 2 | 1.67 | 1.67 | 2 | 2.5

Core • 11-20 • 1.25 | 1.125 | 800 | 0.8 | 0.9 | 1.2 | 1.1 | 1000 | 1.2 | 1.1

Challenge • 21-23 • 500g | 0.1 | 750g

5.3 • Write the Recommendation

OUT OF 22

Warm-Up • 1-12 • 5 | 9 | 10 | 9 | 1 | 6 | 6 | 11 | 12 | 10.02 | 9 | 16

Core • 13-22 • 1 | 52 | 10 | 10 | 52 | 1 | 52 | 8 | 8 | 104

Challenge • 23-26 • 104 | per-unit | 234 | per-unit

Thu • Error Journal Sweep

OUT OF 22

Warm-Up • 1-12 • 0.5 | 0.5 | 2 | 2.4 | 1.5 | 2 | 0.5 | 0.6 | 1.6 | 2.3 | 1.5 | 3

Core • 13-22 • 17.45 | 9.93 | 0.08 | 8 | 3.5 | 19.15 | 9.91 | 0.18 | 7 | 5.6

Challenge • 23-26 • 4.544 | 1 | 5.554 | 1.8

Fri • Mission 04 Test

OUT OF 20

Warm-Up • 1-4 • 3.7 | 0.5 | 3.7 | 0.3

Core • 5-20 • 3.97 | 2.65 | 0.6 | 8 | 12 | 8 | 1.5 | 5.03 | 4.98 | 2.55 | 0.7 | 10.5 | 16 | 7 | 2.25 | 4.04

Challenge • 21-24 • 4.544 | 6 | 5.554 | 7